

Additional Examples and Solutions on Module 5 (System Clock Options)

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Example 1

Compute power and energy consumption during the ON, OFF, and SLEEP modes, according to the following USB meter readings. Then, **compute** the percentage of power and energy savings in the OFF and SLEEP modes compared to the ON mode, assuming the device remains in each mode for 4 s.



LED ON Mode



LED SLEEP Mode



LED OFF Mode

Solution to Example 1

When the LED is in the ON State:

Current, $I_1 = 0.117$ A, Voltage, $V_1 = 5.002$ V and Duration, $t_1 = 4$ s

Power Consumption, $P_1 = V_1 I_1 = 5.002 \times 0.117 = 0.58523$ W

Energy Consumption, $E_1 = P_1 t_1 = 0.58523 \times 4 = 2.341$ J

When the LED is in SLEEP mode:

Current, $I_2 = 0.008$ A, Voltage, $V_2 = 5.069$ V, and Duration, $t_2 = 4$ s

Power Consumption, $P_2 = V_2 I_2 = 5.069 \times 0.008 = 0.04055$ W

Energy Consumption, $E_2 = P_2 t_2 = 0.04055 \times 4 = 0.1622$ J

When the LED is in the OFF State:

Current, $I_3 = 0.019$ A, Voltage, $V_3 = 5.063$ V and Duration, $t_3 = 4$ s

Power Consumption, $P_3 = V_3 I_3 = 5.063 \times 0.019 = 0.0962$ W

Energy Consumption, $E_3 = P_3 t_3 = 0.0962 \times 4 = 0.3848$ J

Percentage of Power Saving in SLEEP mode from LED ON State:

$$\frac{P_1 - P_2}{P_1} \times 100\% = \frac{0.58523 - 0.04055}{0.58523} \times 100\% = 93.07\%.$$

Percentage of Power Saving in LED OFF State from LED ON State:

$$\frac{P_1 - P_3}{P_1} \times 100\% = \frac{0.58523 - 0.0962}{0.58523} \times 100\% = 83.56\%.$$

Percentage of Energy Saving in SLEEP mode from LED ON State:

$$\frac{E_1 - E_2}{E_1} \times 100\% = \frac{2.341 - 0.1622}{2.341} \times 100\% = 93.07\%.$$

Percentage of Energy Saving in LED OFF State from LED ON State:

$$\frac{E_1 - E_3}{E_1} \times 100\% = \frac{2.341 - 0.3848}{2.341} \times 100\% = 83.56\%.$$

Example 2

Determine the output of the following program. **Compute** power and energy consumption during the ON and OFF conditions, as well as SLEEP modes in idle and power down states. **Calculate** power savings in sleep and power down modes in comparison with LED OFF state.

```
#include "LowPower.h"
```

```
void setup() {  
    pinMode(13, OUTPUT);  
}
```

```
void loop() {  
    digitalWrite(13, HIGH);  
    delay(4000);  
    digitalWrite(13, LOW);  
    delay(2000);  
    LowPower.idle(SLEEP_8S, ADC_OFF, SPI_OFF);  
    LowPower.powerDown(SLEEP2S, ADC_OFF, BOD_OFF);  
}
```

// USB Meter readings = 4.971 V, 0.758 A

// USB Meter readings = 5.017 V, 0.125 A

// USB Meter readings = 5.159 V, 0.001 A

// USB Meter readings = 5.089 V, 0.005 A

Solution to Example 2

Initially as pin 13 is set to HIGH, LED will be ON for 4000 ms (4s).
After that, as pin 13 is set to LOW, LED will be OFF for 2000 ms (2s).
Then the device will enter the idle mode and will be in this mode for 8 s.
Finally, the device will be in the power-down mode for 2 s.

Device is in the HIGH state and the LED is ON:

$$I_1 = 0.758 \text{ A}, V_1 = 4.971 \text{ V}, t_1 = 4 \text{ s}$$

$$\text{Power Consumption, } P_1 = V_1 I_1 = 4.971 \times 0.758 = 3.768 \text{ W}$$

$$\text{Energy Consumption, } E_1 = P_1 t_1 = 3.768 \times 4 = 15.072 \text{ J}$$

Device is in the LOW state and the LED is OFF:

$$I_2 = 0.125 \text{ A}, V_2 = 5.017 \text{ V}, t_2 = 2 \text{ s}$$

$$\text{Power Consumption, } P_2 = V_2 I_2 = 5.017 \times 0.125 = 0.627 \text{ W}$$

$$\text{Energy Consumption, } E_2 = P_2 t_2 = 0.627 \times 2 = 1.254 \text{ J}$$

Device is in the idle mode:

$$I_3 = 0.001 \text{ A}, V_3 = 5.15 \text{ V}, t_3 = 8 \text{ s}$$

$$\text{Power Consumption, } P_3 = V_3 I_3 = 5.15 \times 0.001 \text{ A} = 0.00515 \text{ W}$$

$$\text{Energy Consumption, } E_3 = P_3 t_3 = 0.00515 \times 8 = 0.0412 \text{ J}$$

Device is in the low-power power-down mode:

$$I_4 = 0.005 \text{ A}, V_4 = 5.07 \text{ V}, t_4 = 2 \text{ s}$$

$$\text{Power Consumption, } P_4 = V_4 I_4 = 5.08 \times 0.005 \text{ A} = 0.0254 \text{ W}$$

$$\text{Energy Consumption, } E_4 = P_4 t_4 = 0.0254 \times 2 = 0.0508 \text{ J}$$

Power saving during the **idle mode** from the **low state**

$$= (0.627 - 0.00515) / 0.627 \times 100 \% = 99.18\%$$

Power saving during the **low-power power-down mode** from the **low state**

$$= (0.627 - 0.0254) / 0.627 \times 100 \% = 95.97\%$$

Example 3

Compute the total time for which the device is in low-power mode as per the following program if the loop continues for 20 cycles.

```
#include "ArduinoLowPower.h"

void setup() {
  pinMode(LED_BUILTIN, OUTPUT);
}

void loop() {
  digitalWrite(LED_BUILTIN, HIGH);
  delay(1000);
  digitalWrite(LED_BUILTIN, LOW);
  delay(1000);
  LowPower.deepSleep(8000);
}
```

Solution to Example 3

In one cycle, the time for which the device is in low-power deep sleep mode is 8000 ms or 8 s. Hence, if the loop continues for 20 cycles, then the device will be in low-power deep sleep mode for $8 \times 20 = 160$ s.

Example 4

Compute the duration for which the Arduino blinks an LED connected to pin 10 and then enters the power down mode. When powered down, the Arduino's current drops from 10 mA to just 8 μ A. If the Arduino's supply voltage is 5.0 V, then **compute** the amount of power that is saved by the Arduino during this power-down mode.

```
#include "LowPower.h"
```

```
void setup() {  
  pinMode(10, OUTPUT); }
```

```
void loop() {  
  digitalWrite(10, HIGH);  
  delay(1000);  
  digitalWrite(10, LOW);  
  LowPower.powerDown(SLEEP_4S, ADC_OFF, BOD_OFF, SPI_OFF, TWI_OFF);  
  .
```


Solution to Example 4

LED is turned ON for 1000 ms (1 s) and then it is turned OFF and the system enters the power down mode for 4 s. The cycle then repeats. Therefore, the total duration of the cycle is $1 + 4 = 5$ s.

When LED is ON:

Current, $I_{\text{LED ON}} = 10 \text{ mA}$, Voltage, $V = 5 \text{ V}$

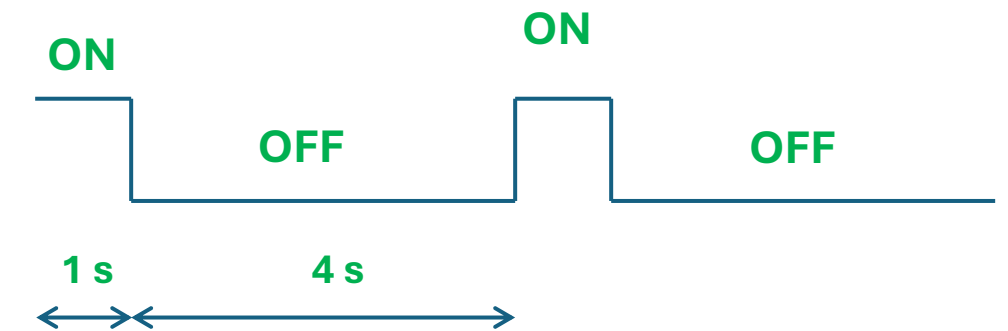
Power Consumption, $P_{\text{LED ON}} = 5\text{V} \times 10\text{mA} = 50 \text{ mW}$

When the device is in power-down mode:

Current, $I_{\text{POWER-DOWN}} = 8\mu\text{A} = 0.008 \text{ mA}$, Voltage, $V = 5 \text{ V}$

Power Consumption, $P_{\text{POWER-DOWN}} = 5 \times 0.008 \text{ mA} = 0.04 \text{ mW}$

Power Saving = $P_{\text{LED ON}} - P_{\text{POWER-DOWN}} = 50 - 0.04 = 49.96 \text{ mW}$



Example 5

Set the Sleep Mode Control Register (SMCR) so that the system remains in power-save mode as a whole and power reduction mode (PRR) for its Timer0, Timer1 and ADC. Also write the setup program in assembly language.

Bit	7	6	5	4	3	2	1	0	
0x33 (0x53)	-	-	-	-	SM2	SM1	SM0	SE	SMCR
Read/Write	R	R	R	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Bit	7	6	5	4	3	2	1	0	
(0x64)	PRTWI	PRTIM2	PRTIM0	-	PRTIM1	PRSPI	PRUSART0	PRADC	PRR
Read/Write	R/W	R/W	R/W	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

SM2	SM1	SM0	Sleep Mode
0	0	0	Idle
0	0	1	ADC Noise Reduction
0	1	0	Power-down
0	1	1	Power-save
1	0	0	Reserved
1	0	1	Reserved
1	1	0	Standby ⁽¹⁾
1	1	1	External Standby ⁽¹⁾

Solution to Example 5

Setup in C:

```
SMCR = 0b00000111;  
PRR  = 0b00101001;
```

Setup in Assembly Language:

```
LDI  R16, 0x07;  
OUT  SMCR, R16;  
LDI  R16, 0x29;  
OUT  PRR, R16;
```

Bit	7	6	5	4	3	2	1	0	
0x33 (0x53)	-	-	-	-	SM2	SM1	SM0	SE	SMCR
Read/Write	R	R	R	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	1	1	1	

Bit	7	6	5	4	3	2	1	0	
(0x64)	PRTWI	PRTIM2	PRTIM0	-	PRTIM1	PRSPI	PRUSART0	PRADC	PRR
Read/Write	R/W	R/W	R/W	R	R/W	R/W	R/W	R/W	
Initial Value	0	0	1	0	1	0	0	1	